Chapter 1

INTRODUCTION

Image forgery, also known as digital image tampering or manipulation, refers to the act of altering or manipulating digital images with the intention to deceive or mislead viewers. With the increasing accessibility of powerful image editing tools and the widespread sharing of digital images on various platforms, image forgery has become a significant concern in today's digital age.

Image forgery can take various forms, including but not limited to copy-move forgery, splicing. Copy-move forgery involves duplicating a region of an image and pasting it within the same image, creating multiple instances of the same content. Splicing occurs when different parts of two or more images are combined to form a new composite image.

The detection of image forgery plays a crucial role in ensuring the integrity and authenticity of digital visual content. It involves developing algorithms and techniques that can identify and localize manipulated regions within an image accurately. Image forgery detection aims to provide forensic evidence and facilitate the identification of tampered images, helping to combat digital fraud, misinformation, and malicious activities.

Detecting image forgery is a challenging task due to the increasing sophistication of image manipulation techniques. As forgers become more adept at concealing their manipulations, the need for advanced and robust forgery detection methods becomes paramount. Researchers and experts in the field of digital image forensics continuously strive to develop innovative approaches to tackle the evolving techniques employed by image forgers.

Recent advancements in machine learning, computer vision, and deep learning have greatly influenced the field of image forgery detection. Techniques such as convolution neural networks (CNNs) and deep learning-based algorithms can be used for detection of various types of image manipulations.

1.1 Problem Definition

The problem of image forgery detection and localization involves developing algorithms and techniques to automatically identify and localize manipulated regions within digital images. The goal is to accurately detect instances of image tampering and provide detailed information about the specific areas that have been altered or manipulated.

Given a digital image, the objective is to determine whether the image has been subjected to any form of forgery, such as copy-move forgery, and splicing. If the image is found to be tampered, the problem further involves localizing and identifying the exact regions that have been manipulated.

Chapter 2

LITERATURESURVEY

2.1 Literature Survey

In (5) the authors focus on the detection and localization of image forgeries using compression fingerprints .The proposed method, called Comprint, leverages the characteristics of compression algorithms to detect manipulated regions in images. Compression fingerprints are obtained by analyzing the compression artifacts introduced by various compression techniques. These fingerprints capture the unique patterns and irregularities caused by image manipulations, such as copy-move, splicing, or content removal.

Comprint consists of two main stages: fingerprint generation and forgery localization. In the fingerprint generation stage, the authors extract features from different compressed versions of an image and construct a set of binary descriptors that represent the compression fingerprints. These descriptors capture the presence or absence of specific compression artifacts. In the forgery localization stage, the authors employ a machine learning approach to classify the image regions as manipulated or authentic based on the extracted fingerprints.

Experimental evaluations on various benchmark datasets demonstrate the effectiveness of Comprint in detecting and localizing image forgeries. The method achieves high accuracy and outperforms several state-of-the-art techniques. Additionally, Comprint exhibits robustness to common image processing operations and can handle JPEG and JPEG2000 compressed images.

The model in (4) consists of two components: the Autoencoder and the one-class SVM. The Autoencoder’s task is to capture the distribution of patches from pristine images. It is coupled with a Discriminator to create a Generative Adversarial Network (GAN). The Discriminator discriminates between patches produced by the Autoencoder and actual patches from pristine images. The architectures of both the Autoencoder and

Discriminator are chosen based on their performance, with the Autoencoder selected based on the lowest mean squared error loss. Various CNN architectures are tested, and the one with the lowest loss is chosen as the best model.

ALGORITHM:

Firstly, a dataset comprising genuine and forged satellite images, along with their corresponding ground truth forgery masks, is collected. The dataset is then preprocessed. Next, the Autoencoder, which is a type of neural network, is trained to capture the distribution of patches from genuine satellite images. The Autoencoder is coupled with a Discriminator, forming a Generative Adversarial Network (GAN). The Discriminator's role is to discriminate between patches produced by the Autoencoder and actual patches from pristine images. The performance of the Autoencoder is evaluated using the mean squared error (MSE) metric, while the Discriminator's performance is measured by binary cross entropy. Multiple CNN architectures are tested for the Autoencoder, and the one with the lowest Mean Square Error (MSE) loss is selected as the best model.

For the forgery detection and localization process, a test satellite image is provided. Patches are extracted from the test image, and these patches are passed through the trained Autoencoder. The MSE loss is calculated for each patch to determine whether it is genuine or forged based on a threshold value. The One-Class Classifier is then utilized to further validate the forgery detection results. Finally, the forged regions are localized by analyzing the detected forged patches and their corresponding locations in the test image.

In (7) the authors focus on the detection of small region forgeries in medical images using a two-stage cascade framework based on Generative Adversarial Networks (GANs). The proposed framework aims to address the challenge of detecting smallscale forgeries in medical images, which can be challenging due to their subtle nature. The first stage of the cascade framework involves utilizing a GAN to generate plausible and diverse patch candidates. The GAN is trained on genuine image patches to learn the underlying distribution. These generated patch candidates are then used to construct a patch candidate set.

In the second stage, a discriminator network is employed to classify the patches from the candidate set as genuine or forged. By comparing the generated patch candidates against the original image, the discriminator network can effectively identify small region forgeries. The two-stage cascade framework enhances the detection accuracy and reduces false positives by combining the generative power of GANs with the discriminative capabilities of the discriminator network.

Chapter 3

PROPOSEDSYSTEM

3.1 System Analysis

DCT model

In (1), the authors propose a model using the statistical features of the DCT co-efficients.

The main concept is to separately calculate the standard deviation and count of non-zero DCT coefficients corresponding to each AC frequency component in order to take advantage of the variance in statistical features of the AC coefficients of the complete picture. The test image and its cropped form are used to assess the recommended characteristics. The SVM classifier is then used to identify the modified/unmodified pictures using the retrieved feature vector.

In the process of pre-processing, a YCbCr colour space conversion is performed on the provided test picture, where Y stands for the luminance component, Cb and Cr for chroma channels. Since most of the picture data in the Y channel of the YCbCr colour model is housed there, human eyes are better able to distinguish between the luminance and chroma components. Due to the sensitivity of human eyesight and the fact that the majority of tampering traces are concealed in the chroma channels, one may mistake a manipulated image for the real thing. Since all three channels Y, Cb, and Cr are exploited in this study, the Final feature is produced by merging all of the extracted features from the Y, Cb, and Cr components.

GAN Model

From (2) the General adversarial networks working and applications have been learned and implemented that concept on image forgery detection. A generative adversarial network (GAN) has two neural networks contest with each other in the form of a zerosum game, where one agent’s gain is another agent’s loss. So for the image forgery detection CASIA1 dataset with 1721 images is used with 800 authentic images that are used for training purpose

Algorithm 1: Classification of Authentic vs Forged Images

Input : Suspicious Image

Output: Classification Result(Authentic/Forged Image (either splicing or Copy-move))

Procedure

Convert RGB Image into YCbCr color space

for *each color sub-image (Y, Cb, Cr)* **do**

  Divide the sub-image into N non-overlapping blocks of size 8×8

  Apply DCT on each block

  for *each block* **do**

    Remove DC coefficient

    Arrange rest of the 63 AC coefficients of block into a column vector :

    $A_i = [a_{2,i}, a_{3,i} \cdots a_{64,i}]^T, \quad 1 \leq i \leq N$

  end

  Concatenate column vector of all N blocks of sub-image to form one matrix of size 63 × N :

$$M_A = [A_1 A_2 \cdots A_N] = \begin{bmatrix} a_{2,1} & a_{2,2} & \cdots & a_{2,N} \\ a_{3,1} & a_{3,2} & \cdots & a_{3,N} \\ \vdots & \vdots & \cdots & \vdots \\ a_{64,1} & a_{64,2} & \cdots & a_{64,N} \end{bmatrix}$$

  Compute row-wise standard deviation s_k and number of ones o_k, $2 \leq k \leq 64$ of matrix M_A for each frequency component

  Form vector of standard deviation, $S = [s_2, s_3 \cdots s_{64}]^T$ and one's, $O = [o_2, o_3 \cdots o_{64}]^T$

  Crop sub-image by removing four rows horizontally and four columns from top left corner

  Form vector of standard deviation, S_C and ones O_C for cropped sub-image

  Concatenate four vectors to obtain feature vector of sub-image :

  $F_{sub-image} = [S^T \quad O^T \quad S_C^T \quad O_C^T]$

end

Combine each of the sub-image feature vector to form final feature vector of the test image :

$F_V = [F_Y \quad F_{Cb} \quad F_{Cr}]$

Apply feature vector F_V to SVM classifier for detection of Authentic/Forged Image

end Procedure

Figure 3.1: Algorithm from the paper [1]

3.1.1 System requirement analysis

3.1.2 Module details of the system

DCT model

The image A x B is divided into K blocks of a particular size, for example, 8x8.A 2D-DCT function is applied on these image blocks. We concentrate only on AC components, so we remove the DC component from the result. These AC components are arranged in a form of a matrix(as shown in Figure 3.1 and then we find the row-wise standard deviation of that matrix. Similarly, No. of ones of each row of coefficients is calculated using the signum function. Then we obtain a cropped version of the original image and calculate all the above-mentioned features to the cropped version too. The difference between the feature vectors defines the classification of the image.

GAN Model

The GAN model consists of two networks generator and the discriminator. Firstly, the generator is fed with both real images and noise as input and fake images are generated by the generator. The discriminator is given real images and the generated images as input and then trained. After training, the fake images from the dataset and real images are fed as input and the discriminator is tested, according to the discriminators output the generator model is updated and both generator and discriminator work as 1 to 1 competition. Generator and discriminator are two neural network models in which generator is deconvolutional network and discriminator is convolutional network. Discriminator model for different epochs is saved and the latest model is used for evaluation.

3.2 System Design

3.2.1 Architecture diagram

GAN Model

Figure 3.2: Architechture diagram for GAN model

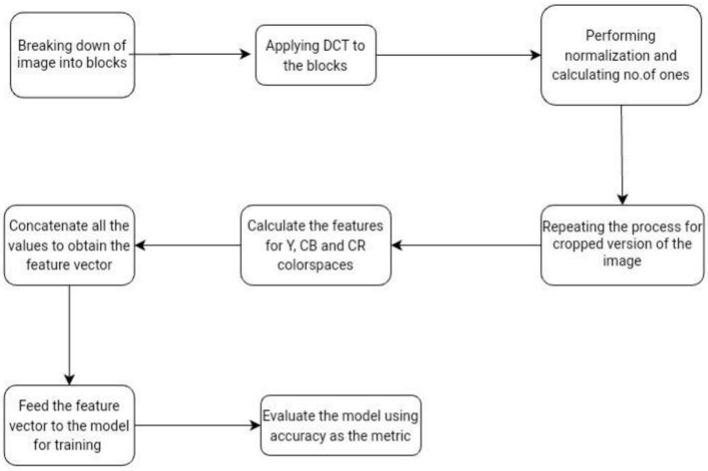

Figure 3.3: Arichitechture diagram of DCT model

Chapter 4

IMPLEMENTATIONANDTESTING

DCT model

The resultant vector in each color space Y,Cb.Cr is obtained after the preprocessing. We merge all these three vectors and finally get the feature vector. We obtain it in the form of a csv file.

We then provide the csv file to the model for training. We use a train test model for this training and testing process. We split the model in the ratio of 8:2 for train and test.Also, we allot different labels to the images while we get the images ready for feature extraction. We label Authentic images as 0, Spliced images as 1 and copy moved images as 2. This helps us in classification of the image as authentic or forged and also if forged, what type of forgery, i.e spliced or copy-move. Then the data is sent for feature extraction and the csv file is obtained along with the assigned label of the image.

To perform image classification, the trained models are utilized. The input is an image along with the trained models, and the output is a classification result indicating whether the image is authentic (0), spliced (1), or copy-move (2), based on the results obtained from the trained models.

GAN Model

For testing the GAN model, the model is been tested with the generated images and real images of CASIA1 dataset as well as CIFAR10 dataset. CIFAR10 dataset has 60000 images and from that 10000 are generated from the generator and fed to discriminator.The GAN model performed better on CIFAR10 because the dataset has a large amount of images so the training of the model is better than that of CASIA1.the generated images are shown in the figure. The accuracy of GAN model on CIFAR10 dataset is
0.65676665. 4.1.

Figure 4.1: Generated images of CASIA1 dataset

Chapter 5

RESULTSANDDISCUSSION

DCT model

In (1) the model using statistical features of DCT coefficients, we used SVM as our primary algorithm to classify the images, we decided to test our model with other algorithms too, in order to understand the basic performances and requirements of our model. We also considered Random Forest classification and KNN along with SVM and compare the results of the performances of all the three models.

We have analysed the above-mentioned algorithm for various block sizes namely 4x4, 8x8, 16x16, 32x32 and 64x64. We have also calculated the difference in performances of same block sizes in two scenarios: Overlapping and Non-Overlapping. Here are the results and model performances of these different blocks.
We infer from 5.1 that when the block sizes are 8x8 and 16x16, the model has performed better than while using other block sizes when they are non-overlapping. We infer from the 5.3 that when the block sizes are 8x8 and 16x16, the model has performed better than while using other block sizes when they are Overlapping

The figure 5.5 shows the predictions from the model and here the 0 stands for authentic image,1 stands for spliced and 2 stands for copy-move. The accuracy of the model is 0.8054054054.. THe confusion matrix in Figure 5.6 is a 3*3 matrix because we have classified into 3 classes .

NON-OVERLAPPING BLOCKS

MODELS	Accuracy for 4x4	Accuracy for 8x8	Accuracy for 16x16	Accuracy for 32x32	Accuracy for 64x64
SVM	0.88	0.96	0.95	0.92	0.86
RANDOM FOREST	0.73	0.85	0.96	0.84	0.80
KNN	0.64	0.74	0.72	0.67	0.61

Figure 5.1: Accuracy for non-overlapping blocks

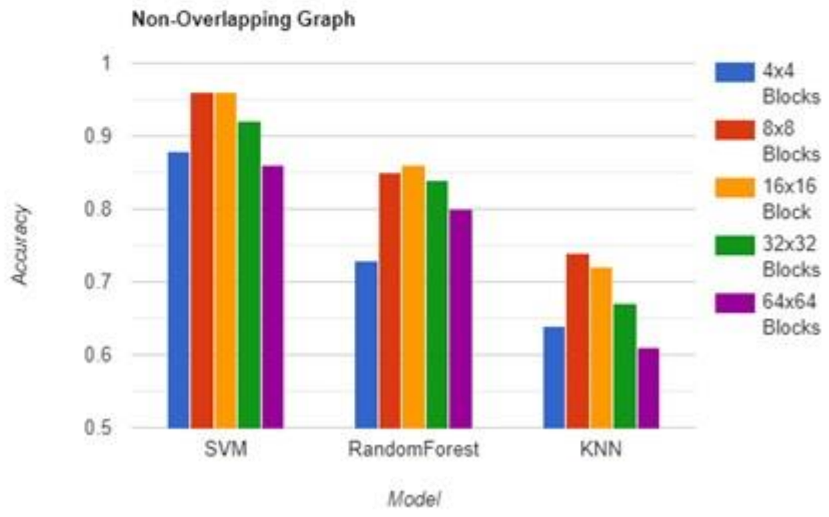

Figure 5.2: Accuracies for different non-overlapping block sizes

OVERLAPPING BLOCKS

MODELS	Accuracy for 4x4	Accuracy for 8x8	Accuracy for 16x16	Accuracy for 32x32	Accuracy for 64x64
SVM	0.85	0.94	0.97	0.95	0.54
RANDOM FOREST	0.71	0.84	0.84	0.84	0.81
KNN	0.62	0.71	0.71	0.68	0.64

Figure 5.3: Accuracy for overlapping blocks

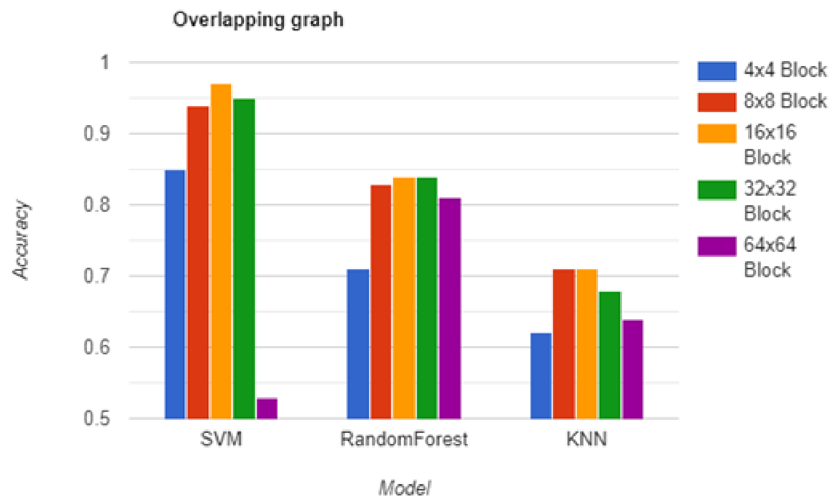

Figure 5.4: Accuracies for different overlapping block sizes

	Authentic	Spliced	Copy move
Authentic	154	5	3
Spliced	11	60	15
Copy move	3	16	79

Figure 5.6: Confusion matrix for DCT model

```
predictions

array([0, 2, 0, 1, 2, 0, 1, 1, 0, 0, 1, 0, 1, 0, 0, 2, 2, 0, 2, 0, 1, 0,
       1, 0, 0, 0, 2, 0, 0, 0, 0, 1, 1, 0, 2, 0, 1, 0, 2, 0, 0, 0, 0, 2,
       0, 0, 0, 1, 2, 2, 2, 1, 0, 2, 0, 1, 0, 1, 2, 2, 1, 1, 0, 0, 0, 0,
       0, 1, 0, 2, 0, 2, 1, 1, 0, 0, 0, 0, 0, 2, 1, 1, 2, 1, 2, 0, 1, 2,
       0, 2, 0, 0, 0, 0, 0, 2, 1, 0, 0, 2, 1, 2, 1, 0, 0, 0, 0, 2, 0, 1,
       1, 2, 0, 0, 1, 2, 0, 2, 0, 0, 2, 1, 0, 1, 0, 0, 1, 0, 0, 0, 2, 2,
       0, 0, 2, 0, 2, 2, 2, 2, 0, 0, 2, 0, 0, 2, 2, 2, 2, 2, 2, 1, 0, 0,
       0, 0, 2, 0, 2, 2, 2, 0, 0, 0, 1, 0, 2, 1, 2, 2, 1, 0, 0, 2, 1, 0,
       0, 2, 1, 1, 0, 0, 1, 0, 0, 2, 0, 0, 2, 1, 2, 1, 0, 1, 0, 1, 0, 1,
       0, 0, 0, 2, 2, 1, 0, 0, 0, 0, 0, 1, 0, 0, 2, 0, 1, 0, 1, 0, 1, 2,
       1, 0, 0, 0, 2, 0, 2, 2, 2, 2, 0, 1, 2, 0, 2, 1, 0, 0, 0, 2, 0, 0,
       2, 2, 2, 1, 0, 2, 2, 1, 0, 1, 0, 0, 0, 2, 0, 0, 1, 0, 1, 0, 2, 2,
       0, 0, 0, 0, 2, 2, 0, 2, 1, 0, 0, 0, 1, 0, 0, 1, 2, 1, 0, 0, 0, 0,
       1, 1, 0, 1, 2, 0, 2, 0, 2, 0, 0, 2, 1, 0, 0, 0, 2, 0, 1, 1, 1, 2,
       2, 0, 0, 0, 2, 2, 0, 0, 2, 1, 2, 1, 0, 2, 0, 1, 2, 1, 1, 1, 2, 0,
       1, 2, 2, 1, 1, 1, 0, 0, 0, 0, 1, 0, 0, 0, 1, 2])
```

Figure 5.5: Predictions made in DCT model indicating what kind of forgery

GAN Model

The confusion matrix in Figure 5.7 has more number of true negatives. Here the true negatives are the real images detected as real. so some changes in the model will improve the accuracy and the model will be more accurate in predicting the forgeries. The accuracy,precision,recall,f1 score of the GAN model are

Accuracy of the Discriminator: 0.60158

Precision of the Discriminator: 0.63791

Recall of the Discriminator: 0.87762

F1-Score of the Discriminator: 0.73881

CONFUSION MATRIX	**Positive (1)**	**Negative (0)**
Positive (1)	**48** (TP)	**99** (FP)
Negative (0)	**403** (FN)	**710** (TN)

Figure 5.7: Confusion matrix for GAN model

```
Discriminator output for image  1 : 0.69359493
1/1 [==============================] - 0s 66ms/step
Discriminator output for image  2 : 0.38000697
1/1 [==============================] - 0s 47ms/step
Discriminator output for image  3 : 0.36338946
1/1 [==============================] - 0s 50ms/step
Discriminator output for image  4 : 0.8277403
1/1 [==============================] - 0s 64ms/step
Discriminator output for image  5 : 0.80629885
1/1 [==============================] - 0s 56ms/step
Discriminator output for image  6 : 0.85121745
1/1 [==============================] - 0s 51ms/step
Discriminator output for image  7 : 0.84380865
1/1 [==============================] - 0s 60ms/step
Discriminator output for image  8 : 0.87073874
1/1 [==============================] - 0s 47ms/step
Discriminator output for image  9 : 0.660874
1/1 [==============================] - 0s 54ms/step
Discriminator output for image  10 : 0.25385526
```

Figure 5.8: Discriminator output for CASIA1 dataset. (> 0.5- Real image, < 0.5 -fake image

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